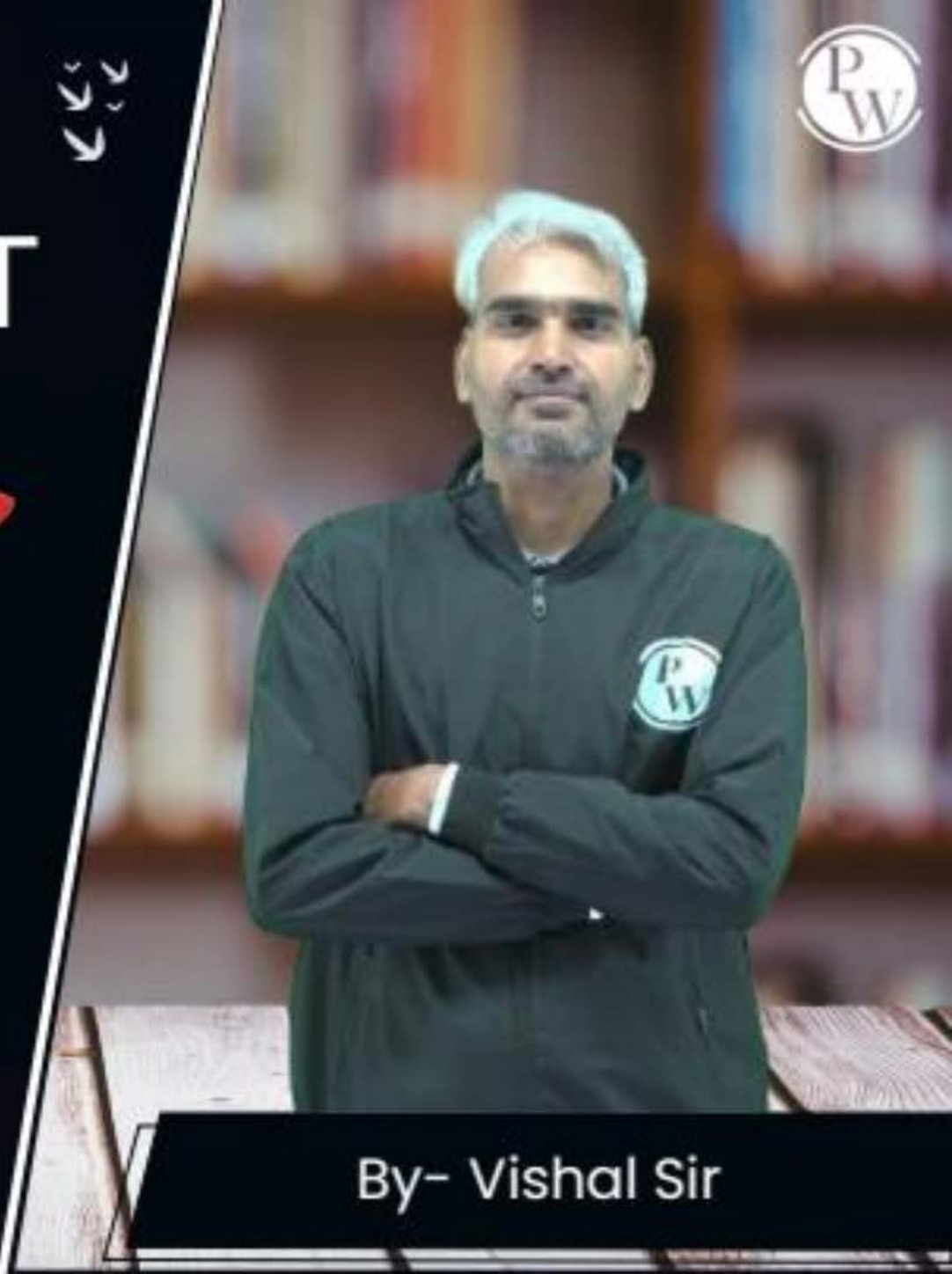
Computer Science & IT



Database Management System

Query Languages

Lecture No. 09



By- Vishal Sir

Recap of Previous Lecture

- Topic: Practice questions
- Topic: AS clause



Slide

Topics to be Covered



- Topic: AS clause
- Topic: WITH clause
- Topic: Comparison with NULL
- Topic: Comparison with regular expression

Slide



Topic : AS clause

AS clause is used to rename a column or table with an alias. {Note: AS Clause can be used to give a new_name (alias) to almost Everything}

- An alias only exists for the duration of the query.

#e.g. Consider a database that has the relation schema EMP (EmpId, EmpName, [redacted] DeptName).
An instance of the schema EMP and a SQL query on it are given below:

EmpId	Emp Name	DeptName
1	XYA	AA
2	XYB	AA
3	XYC	AA
4	XYD	AA
5	XYE	AB
6	XYF	AB
7	XYG	AB
8	XYH	AC
9	XYI	AC
10	XYJ	AC
11	XYK	AD
12	XYL	AD
13	XYM	AE

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
    (SELECT DeptName, COUNT(EmpId) AS EC(DeptName, Num)
                                    FROM EMP
                                    GROUP BY DeptName)
```

The output of executing the SQL query is_____

#e.g. Consider a database that has the relation schema EMP (EmpId, EmpName, and DeptName).
An instance of the schema EMP and a SQL query on it are given below:

Outer query

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
    (SELECT DeptName, COUNT(EmpId) AS EC(DeptName, Num)
                                    FROM EMP
                                    GROUP BY DeptName)
```

Select DeptName, Count(EmpId) AS EC(DeptName, Num)

EC

DeptName	Num
DeptName	Count(EmpID)

o/p =

The output of executing the SQL query is _____

inner query

#e.g. Consider a database that has the relation schema
EMP (EmpId, EmpName, and DeptName).
An instance of the schema EMP and a SQL query on it are given below:

Outer query

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
    (SELECT DeptName, COUNT(EmpId) AS EC(DeptName, Num)
                                    FROM EMP
                                    GROUP BY DeptName)
```

The output of executing the SQL query is _____

inner query

↓

it is independent inner query

∴ inner query will be executed first,

Because of execution of inner query relation EC (DeptName, Num) will come into the picture

It will exist only during the execution of overall query

#e.g. Consider a database that has the relation schema EMP (EmpId, EmpName, [redacted] DeptName).
An instance of the schema EMP and a SQL query on it are given below:

EmpId	Emp Name	DeptName
1	XYA	AA
2	XYB	AA
3	XYC	AA
4	XYD	AA
5	XYE	AB
6	XYF	AB
7	XYG	AB
8	XYH	AC
9	XYI	AC
10	XYJ	AC
11	XYK	AD
12	XYL	AD
13	XYM	AE

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
(SELECT DeptName, COUNT(EmpId) AS EC(DeptName, Num)
 FROM EMP
 GROUP BY DeptName)
```

The output of executing the SQL query is _____

#e.g. Consider a database that has the relation schema EMP (EmpId, EmpName, and DeptName).
An instance of the schema EMP and a SQL query on it are given below:

```
SELECT AVG(EC.Num)
FROM EC                                          (4)
WHERE (DeptName, Num) IN
(SELECT DeptName, COUNT(EmpId) AS EC(DeptName, Num)   (3)
 FROM EMP                                        (1)
 GROUP BY DeptName)                              (2)
```

The output of executing the SQL query is _____

output of inner query

EC

DeptName	Num
AA	4
AB	3
AC	3
AD	2
AE	1

o/p of select

→ after execution of inner query, we will execute outer query

DeptName	Num
AA	4
AB	3
AC	3
AD	2
AE	1

for all tuples; where Condⁿ is true

$\Rightarrow$ $Avg(Num) = \frac{4+3+3+2+1}{5}$

$= \frac{13}{5} = 2.6$



Topic : WITH clause

'With' Clause is used along with 'AS'

- The WITH Clause is mainly used to provide a subquery block a name that can be referenced by the main SQL query or the subqueries that follows.

Syntax of WITH :-

Here we will define a new schema

new names assigned to the columns

output of this sub query will be recognized by a new-schema defined by 'WITH' clause

```
WITH  New-rel^n name(nA1, nA2 . . . .)  AS  Sub-query
```

We may have multiple Sub-queries using WITH clause

In the end we will have Main query

```
Select . . . .
. . . . .
. . . . .
```

Main query

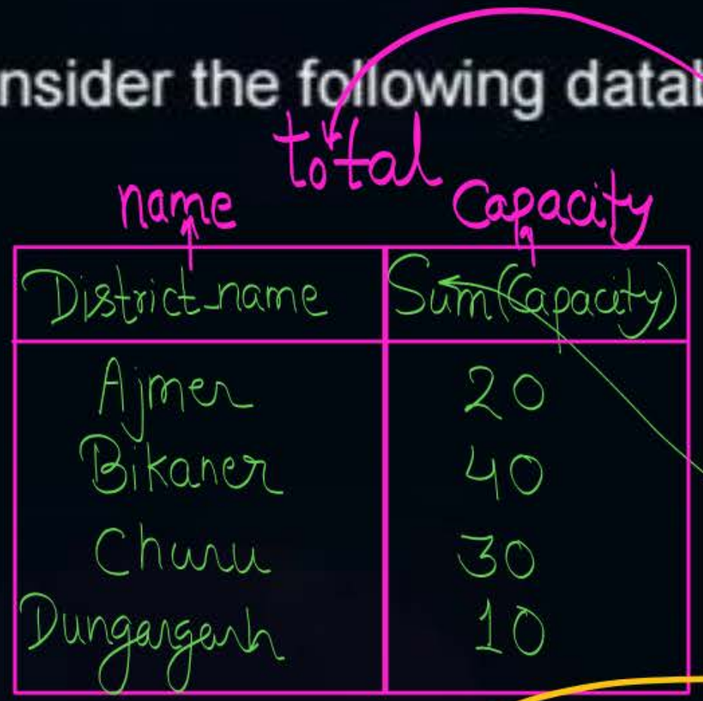
#e.g. Consider the following database table named water_scheme

water_scheme		
scheme_no	district_name	capacity
1	Ajmer	20
1	Bikaner	10
2	Bikaner	10
3	Bikaner	20
1	Churu	10
2	Churu	20
1	Dungargarh	10

The number of tuples returned by the following SQL query is_____

```
with total(name, capacity) as
        select district_name, sum(capacity)
        from water_schemes
        group by district_name
with total_avg(capacity) as
        select avg(capacity)
        from total
select name from total, total_avg
        where total.capacity >= total_avg.capacity
```

#e.g. Consider the following database table named water_scheme

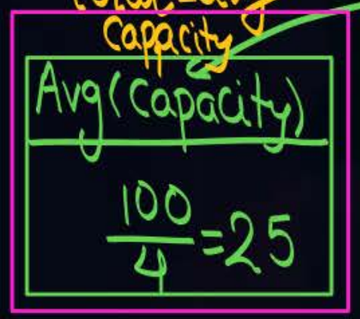


The number of tuples returned by the following SQL query is_____

```
with total(name, capacity) as
        select district_name, sum(capacity)
        from water_schemes
        group by district_name
with total_avg(capacity) as
        select avg(capacity)
        from total
select name from total, total_avg
        where total.capacity >= total_avg.capacity
```

new schema

Sub-query



#e.g. Consider the following database table named water_scheme

District_name	Sum(Capacity)
Ajmer	20
Bikaner	40
Churu	30
Dungargarh	10

The number of tuples returned by the following SQL query is_____

```
with total(name, capacity) as
        select district_name, sum(capacity)
        from water_schemes
        group by district_name
with total_avg(capacity) as
        select avg(capacity)
        from total
select name from total, total_avg
        where total.capacity >= total_avg.capacity
```

Avg(capacity)
$\frac{100}{4}=25$

total.name	total.Capacity	total_avg.Capacity
Ajmer	20	25
Bikaner	40	25
Churu	30	25
Dungargarh	10	25

name
Bikaner
Churu

#Q. Consider the relational database with the following four schemes and their respective instances.

Student(sNo, sName, dNo) Dept(dNo, dName)
Course(cNo, cName, dNo) Register(sNo, cNo)

Student

sNo	sName	dNo
S01	James	D01
S02	Rocky	D01
S03	Jackson	D02
S04	Jane	D01
S05	Milli	D02

Dept

dNo	dName
D01	CSE
D02	EEE

Course

cNo	cName	dNo
C11	DS	D01
C12	OS	D01
C21	DE	D02
C22	PT	D02
C23	CV	D03

Register

sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

Question Continues in Next Slide

[NAT: 2022: 2M]

#Q. SQL Query:

```
SELECT * FROM Student AS S
WHERE NOT EXIST
(SELECT cNo FROM Course WHERE dNo = "D01"
EXCEPT
SELECT cNo FROM Register WHERE sNo = S.sNo)
```

outer query

inner query

independent

CNO
C11
C12

(Minus)

correlated

The number of rows returned by the above SQL query is _____.

This query will produce "CNO" of all the Courses for which any specific student has "registered"

#Q. Consider the relational database with the following four schemes and their respective instances.

Student(sNo, sName, dNo) Dept(dNo, dName)
Course(cNo, cName, dNo) Register(sNo, cNo)

Student (S)

sNo	sName	dNo
S01	James	D01
S02	Rocky	D01
S03	Jackson	D02
S04	Jane	D01
S05	Milli	D02

Register

sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

Question Continues in Next Slide

#Q. Consider the relational database with the following four schemes and their respective instances.

Student(sNo, sName, dNo) Dept(dNo, dName)
Course(cNo, cName, dNo) Register(sNo, cNo)

Student

sNo	sName	dNo
S01	James	D01
S02	Rocky	D01
S03	Jackson	D02
S04	Jane	D01
S05	Milli	D02

Register

sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

Question Continues in Next Slide

#Q. Consider the relational database with the following four schemes and their respective instances.

Student(sNo, sName, dNo) Dept(dNo, dName)
Course(cNo, cName, dNo) Register(sNo, cNo)

Student		
sNo	sName	dNo
S01	James	D01
S02	Rocky	D01
S03	Jackson	D02
S04	Jane	D01
S05	Milli	D02

Register	
sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

Question Continues in Next Slide

#Q. Consider the relational database with the following four schemes and their respective instances.

Student(sNo, sName, dNo) Dept(dNo, dName)
Course(cNo, cName, dNo) Register(sNo, cNo)

Student		
sNo	sName	dNo
S01	James	D01
S02	Rocky	D01
S03	Jackson	D02
S04	Jane	D01
S05	Milli	D02

Register	
sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

Question Continues in Next Slide

#Q. Consider the relational database with the following four schemes and their respective instances.

Student(sNo, sName, dNo) Dept(dNo, dName)
Course(cNo, cName, dNo) Register(sNo, cNo)

Student

sNo	sName	dNo
S01	James	D01
S02	Rocky	D01
S03	Jackson	D02
S04	Jane	D01
S05	Milli	D02

Register

sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

Question Continues in Next Slide

#Q. Consider the relational database with the following four schemes and their respective instances.

Student(sNo, sName, dNo) Dept(dNo, dName)
Course(cNo, cName, dNo) Register(sNo, cNo)

Student (S)		
sNo	sName	dNo
S01	James	D01
S02	Rocky	D01
S03	Jackson	D02
S04	Jane	D01
S05	Milli	D02

"cNo" of the courses for which corresponding student has "registered"

cNo
{C11, C12}
{C11}
{C21, C22, C23}
{C11, C12}
{C11, C21}

Register	
sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

Question Continues in Next Slide

[NAT: 2022: 2M]

#Q. SQL Query:

```
SELECT * FROM Student AS S
WHERE NOT EXIST
    (SELECT cNo FROM Course WHERE dNo = "D01"
        EXCEPT
    SELECT cNo FROM Register WHERE sNo = S.sNo)
```

The number of rows returned by the above SQL query is _____

Student

SNO	Sname	dNO
S01		
S02		
S03		
S04		
S05		

cNO	−	CNO	
C11, C12	−	C11, C12	∅
C11, C12	−	C11	C12
C11, C12	−	C21, C22, C23	C11, C12
C11, C12	−	C11, C12	∅
C11, C12	−	C11, C21	C12

Not exists will return True

∴ Records of 'S01' & 'S04' are selected



Topic : Comparison with NULL

* NULL is a non-zero unknown value. and no two NULL are equal

Note :- To Compare with NULL, operator used is "IS"

Complement of "IS" is "IS NOT"

Query: Retrieve Eids of the employee with no passport.

EMP

Eid	Ename	Pass_NO
E_1	Amit	12345
E_2	Ram	NULL
E_3	Anand	27935
E_4	Anmol	85834
E_5	Ramanand	76349

```
Select Eid
From EMP
Where Pass_NO = NULL
```

Wrong, because NO two NULL are equal

```
Select Eid
From EMP
Where Pass_NO = 0
```

Wrong, because NULL is non-zero

Correct query:-

```
Select Eid
From EMP
Where Pass_NO IS NULL
```

EMP

Eid	Ename	Pass_NO
E_1	Amit	12345
E_2	Ram	NULL
E_3	Anand	27935
E_4	Anmol	85834
E_5	Ramanand	76349

Query: Retrieve Eids of the employee with a valid Passport

```
Select Eid
From EMP
Where Pass_NO IS NOT NULL
```

H.W.

#Q. Consider The Following Relational Scheme

Student (school-id, sch-roll-no, sname, saddress)
School (school-id, sch-name, sch-address, sch-phone)
Enrolment (school-id, sch-roll-no, erollno, examname)
ExamResult (Erollno, examname, marks)

(A)for each school with more than 200 students appearing in exams, the name of the school and the number of 100s scored by its students

(B)for each school with more than 200 students in it, the name of the school and the number of 100s scored by its students

(C)for each school with more than 200 students in it, the name of the school and the number of its students scoring 100 in at least one exam

(D)nothing; the query has a syntax error

Question Continues in Next Slide

H.W.

[MCQ: 2008: 2M]

#Q. What does the following SQL query output?

```
SELECT sch-name, COUNT (*)
FROM School C, Enrolment E, ExamResult R
WHERE     E.school-id = C.school-id
                    AND
          E.examname = R.examname
                    AND
          E.erollno = R.erollno
                    AND
          R.marks = 100
                    AND
          E.school-id IN
                    (SELECT school-id
                    FROM student GROUP BY school-id HAVING COUNT (*) > 200)
GROUP By school-id
```

H.W: (Just Provide your answer, None of the given option may be Correct)

#Q. Consider the following relational schema: [MCQ: 2009: 2M]

Suppliers(sid:integer, sname:string, city:string, street:string)
Parts(pid:integer, pname:string, color:string)
Catalog(sid:integer, pid:integer, cost:real)

Consider the following relational query on the above database:

```
SELECT S.sname
        FROM Suppliers S
        WHERE S.sid NOT IN
                (SELECT C.sid
                        FROM Catalog C
                        WHERE C.pid NOT IN
                                (SELECT P.pid
                                        FROM Parts P
                                        WHERE P.color<> \'blue\'))
```

Question Continues in Next Slide

H.W. (

#Q. Assume that relations corresponding to the above schema are not empty. Which one of the following is the correct interpretation of the above query?

A. Find the names of all suppliers who have supplied a non-blue part.

B. Find the names of all suppliers who have not supplied a non-blue part.

C. Find the names of all suppliers who have supplied only blue parts.

D. Find the names of all suppliers who have not supplied only blue parts.



2 mins Summary

- **Topic** AS clause
- **Topic** WITH clause
- **Topic** Comparison with NULL
- **Topic** Comparison with regular expression

Slide

if $X \rightarrow\rightarrow Y$ & $X \rightarrow\rightarrow Z$, then
$X \rightarrow\rightarrow YZ$ will hold true

THANK - YOU

But, if $X \rightarrow\rightarrow YZ$
then $X \rightarrow\rightarrow Y$ and $X \rightarrow\rightarrow Z$ need not hold true

Slide